Mark Colley

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RESEARCH INTERESTS

My multidisciplinary research in Human-Computer Interaction (HCI), Accessibility, & Computational Methods is dedicated to tackling complex challenges and seizing opportunities within advanced mobility technologies. It involves designing, implementing, & testing novel simulators to study futuristic mobility scenarios. My work aims to address issues such as undertrust in automated vehicles and to enhance accessibility in urban (air) mobility, thereby supporting societal and industrial growth. A significant portion of my research focuses on evaluating innovative interaction paradigms between automated vehicles and vulnerable road users, utilizing empirical evidence alongside simulation-based approaches to analyze their broad-scale impacts.

EDUCATION

Ulm University — Ulm, Germany 04/2019 — 26.04.2024
Doctor of Science (PhD) in Human–Computer Interaction
Thesis Title: *Calibrating Trust in Automated Vehicles-Theoretical, Design, & Empirical Insights into Effects of Visualizations on Trust*
Advisor: Prof. Dr. Enrico Rukzio
Committee: Prof. Dr. Stephen Brewster (University of Glasgow, UK), Prof. Dr. Wendy Ju (Cornell Tech, NYC, USA)

Cornell Tech — New York, USA 01/2023 — 04/2023 & 07/2023 — 08/2023
Visiting Research Scholar with Prof. Dr. Wendy Ju
Note: supported by the German Academic Exchange Service (DAAD)

Ulm University — Ulm, Germany 10/2015 — 11/2018
Master of Science in Computer Science — Overall Grade: 1.1 - A-equivalent
Thesis Title: *Identification, Investigation and Classification of Use Cases for Cooperation in Highly Autonomous Driving and the Applicability thereof* — Grade: 1.0 - A-equivalent

DHBW Ravensburg, Campus Friedrichshafen — Friedrichshafen, Germany 09/2012 — 10/2015
Bachelor of Engineering: Information Technology
Note: cooperative degree as an employee of Airbus Defence and Space GmbH

JOB EXPERIENCE

University College London — London, United Kingdom
Lecturer/Assistant Professor — 10/2024 — present
As the coordinator of the UCLIC Research Seminar Series, I organize and manage a vibrant seminar program featuring leading researchers in HCI. The recorded seminars are accessible through the UCLIC YouTube Channel. I also lead two modules: PSYC0095 (Future Interfaces), which explores emerging technologies, & PSYC0097 (Interaction Design), focusing on design processes, theory, & practical application. In my lab management role, I maintain an active and collaborative research environment that facilitates cutting-edge HCI work at UCLIC.

University of Tokyo — Tokyo, Japan
Visiting Professor — 04/2025 — 06/2025 & 12/2025 — 02/2026
Supported by Canon, I am a visiting Professor in the User Interface Research Group led by Takeo Igarashi.

Ulm University — Ulm, Germany
Research Associate — 04/2019 — 09/2024
I significantly contributed to research and teaching and played a key role in securing and implementing three externally funded projects—INTUITIVER, SituWare, & SEMULIN. Furthermore, I successfully championed novel topics related to accessibility and computational methods into the department's research scope. My work incorporates numerous research approaches such as experiments, interviews, focus groups, literature reviews, observation, usability testing, & dataset creation. It has led to over **30 peer-reviewed first-author full papers**. Since May 2024, I have been working as a postdoctoral researcher.

Airbus Defence and Space GmbH — Ulm, Germany
Software Engineer — 09/2012 — 03/2019
I developed realistic simulations for aviation use cases, ensuring they closely mirror real-world conditions. My work included thorough testing to guarantee software reliability and performance, coupled with a comprehensive analysis of project requirements to create efficient and client-focused solutions. Additionally, I deployed and integrated these solutions, ensuring a smooth transition and optimal functionality within a safety-critical system that has existed and is being enhanced for >20 years. In 2014, I spent **3 months at Airbus Group in Newport, Wales**, contributing to system engineering tasks.

Ulm University — Ulm, Germany
Student Research Assistant — 01/2019 — 03/2019
Led comprehensive literature reviews and significantly contributed to developing both a theoretical and an empirical publication on human-vehicle cooperation. My role involved critically analyzing existing works, synthesizing key findings, & collaboratively shaping the direction of the publication. Additionally, I actively participated in internal review processes.

GRANTS

- 2024 V&A Creative Experience Futures Lab (Co-Investigator, 10.000£)
- 2024 Cornell–UCL Global Strategic Collaboration Award: "Evaluating Cultural Differences in Mobility: Developing a Toolkit and Preliminary Investigations" (Co-Principal Investigator, 10.000$)
- Google PaliGemma Academic Program (5.000$)

MEMBERSHIPS, ACADEMIC AND INDUSTRIAL SUPPORT

- Recipient of the Canon Foundation Research Fellowship 2025 (7.500€)
- Recipient of the Deutscher Akademischer Austauschdienst (DAAD, German Academic Exchange Service) research stipend 2022 for my research visit at Cornell Tech
- Startup your career support (10.000€) by Ulm University
- Hardware support by Leia Inc. and NextMind (acquired by Snap)
- Heidelberg Laureate Forum Foundation alumnus, AlumNode & HLFF Inspiring Minds member, mentor: Gabriele Kotsis
- Co-Founder of Zefwih since 01/2022, supported by the German Federal Ministry of Transport and Digital Infrastructure
- Affiliate member of the United Kingdom Advising and Tutoring (UKAT) association since 11/2024

SERVICE AND VOLUNTEERING ACTIVITIES

- Reviewing of Project Proposals:
  - Linz Institute of Technology (LIT) Seed Funding Program, JKU, Austria, 2024
  - NWO - Open Competition Domain Science-M, 2024
  - Member of the EPSRC Peer Review College, joined 11/2024
- Associate Chair / Program Committee Member:
  - Subcommittee Chair for the Full Paper track at AutoUI '25
  - Associate Editor for **IMWUT**, **Behaviour & Information Technology**, & **IJHCS**
  - Full Paper - **CHI '24'25**, **CSCW '24**, AutoUI '21-'24, ETRA '24'25, MobileHCI '23'24'25, MuC '22'23'24, CUI '24'25, CHIWORK'25, ISMAR'25
  - Member of the **Best Paper Committee** for **CHI '24**
  - Late-Breaking Works (LBW) - CHI '22'-25, TEI '23'24, CHI Play '24
- **Organizing Committee**: AutoUI 2023 (Workshop Chair), AutoUI 2024 (Demo Chair), AutoUI 2025 (Workshop & Tutorial Chair), MuC 2025 Workshop & Tutorial Chair
- Peer Reviewing: **Over 500 peer-reviews completed** so far for **CHI**, **TVCG**, MobileHCI, UIST, CUI, DIS, TEI, IUI, ISMAR, **IJHCS**, TRF, ETRA, AutoUI, CHI Play, ICMI, EICS, VRST, IEEE VR, **CSCW**, IJHCI, IMWUT, ESWA, **ToCHI**; Reviewer for project ERROR
- Examiner appointment (11/2024) University of Ulm in the courses: Media Informatics, Computer Science, Software Engineering, and AI - graded 5+ Master theses
- Student Volunteer: UbiComp 2022
- Participation in **ISO standardization process** regarding driving simulators
- **Main organizer** of the Post-CHI Summer School On Automotive User Interfaces and Future Mobility
- Co-Organizer of the German Pre-CHI 2022 hosted in Ulm, Germany with more than 70 participants
- **Local council** (2019 — 2024) in 89155 Erbach-Donaurieden, Germany
- Former Lifeguard with the Deutsche Lebens-Rettungs-Gesellschaft e.V. (DLRG; German Life Saving Association)

TEACHING

Future Interfaces (PSYC0095)
Course Organizer: Co-organization of the interdisciplinary course, emphasizing user-centered design and design thinking with an innovative approach to HCI. Spring 2025

Research Project in Human–Computer Interaction
Course Organizer: Co-organization of the interdisciplinary project, emphasizing user-centered design and design thinking, integrated with a year-long, research-driven group project that culminated in several publications. Fall 2019 — Fall 2023

Research Trends in Media Informatics
Course Organizer: Co-organization of the course, mentoring PhD students on course structure and content, & personally delivering in-depth, one-on-one instruction to over 15 students on conducting literature surveys using the PRISMA method, complemented by active involvement in student assessment and grading processes. Fall 2019 — Fall 2023

Automotive user interfaces and interactive vehicle applications
Course Organizer and Lecturer: Actively participated in the co-development of comprehensive course materials, aligning them with industry standards and academic requirements. Delivered an in-depth lecture focused on external communication of automated vehicles, a key component crucial for students' success in the final assessment. Contributed to the evaluation process by assisting in grading, ensuring a fair and thorough assessment of student performance. Fall 2022

PHD MENTORSHIP

I have the privilege to be working with and mentoring 8 PhD students and candidates.

Active

Ulm University
- Pascal Jansen (since 2021)
- Annika Stampf (since 2021)
- Luca-Maxim Meinhardt (since 2022)
- Max Rädler (since 2024)
- Julian Britten (since 2025)

TH Ingolstadt
- Mathias Haimerl (since 2021)

UCL
- Lan Xiao (since 2024)
- Coral Zawadzki (since 2025)

THESIS SUPERVISION (Main Supervisor, Selection, Total>60)

Bachelor theses (all Ulm University)
- Simon Kopp (2024)
- Jonathan Westhauser (2023)
- Alexander Fassbender (2022)
- Julian Britten (2022; **now PhD student at Ulm University**)
- Elvedin Bajrovic (2021)
- Tim Fabian (2021)
- Max Rädler (2021)
- Svenja Krauß (2020)
- Jan Henry Belz (2020; **now PhD student at Ulm University in collab. with Porsche AG**)
- Stefanos Mytilineos (2020)
- Christian Bräuner (2019)

Master theses (all Ulm University)
- Marcel Giss (2025)
- Tim Fabian (2025)
- Julian Britten (2025; **now PhD student at Ulm University**)
- Svenja Krauß (2023)
- Max Rädler (2023; **now PhD student at Ulm University**)
- Bastian Wankmüller (2022)
- Cristina Evangelista (Ulm University and Cerence GmbH; 2022)
- Albin Zeqiri (2022; **now PhD student at Ulm University**)
- Annika Stampf (Ulm University and Mercedes Benz AG; 2021; **now PhD student at Ulm University**)
- Pascal Jansen (2021; **now PhD student at Ulm University**)
- Leon Dönch (University of Munich; 2020); co-supervised with Dr. Kai Holländer
- Gülsemin Dogru (Ulm University and Bosch GmbH; 2019)

INVITED TALKS AND SEMINARS (Selection)

Talks
- Miraikan – The National Museum of Emerging Science and Innovation (16.05.2025); in-person
- University of Tokyo User Interface Research Group (09.05.2025): "Shaping the Future of Mobility"; in-person
- University of Cambridge Department of Engineering (31.03.2025); in-person
- Birmingham City University Research Centre (17.12.2024); in-person
- University of Sydney Design Lab (09.08.2024); in-person
- UC Calgary iLab (01.08.2023): "Automotive UI: The Intersection of Accessibility, Computational Methods, & Design", online
- KIT at Human-Centered Systems Lab (01.12.2022): "Inclusive Interaction with Future Mobility", in-person

Seminars
- Decision & Design Spring Retreat (02.04.2025 - 04.04.2025; Guest speaker): "Shaping the Future of Mobility"; in-person
- Heidelberg Laureate Forum (23.09 - 30.09.2023)
- Dagstuhl Seminar - Radical Innovation and Design for Connected and Automated Vehicles (May 29 – Jun 03, 2022)

AWARDS

ACM SIGCHI Emerging Researcher Recognition 2025: I am honored to have been recognized by the ACM SIGCHI *for early-career contributions to inclusive HCI research, extensive community building, and championing open science in automotive user interfaces.*

Outstanding Reviewer Recognition: AutoUI '21, CHI '21, CHI '22, 4xCHI '24, 2xCHI'25, IMWUT '23, MobileHCI '23'24 (3x), UIST '23, CHI Play '24 (3x)
I have been honored with eighteen Outstanding Reviewer Awards and was named **Distinguished Reviewer** for MobileHCI'23, showing my commitment to excellence in academic review processes. These accolades reflect my deep understanding and critical analysis skills, contributing significantly to the advancement of scholarly discourse.

Honorable Mention Award at CHI'25 - DOI: 10.1145/3706598.3713514
This paper addressed the challenge in scalable automotive user interface design. We implemented *OptiCarVis*, a system integrating Human-in-the-Loop Bayesian Optimization. An online study (N=117) demonstrates OptiCarVis efficacy in significantly improving trust,

acceptance, perceived safety, and predictability without increasing cognitive load.

Honorable Mention Award at CHI'24 - DOI: 10.1145/3613904.3642341
This paper addressed the challenge in automotive user interface design where testing transitions from lab-based driving simulators to on-road studies to increase ecological validity faces difficulties in replicating studies. A platform-portable infrastructure named *Portobello* facilitates running identical studies in-lab and on-road. A proof-of-concept was demonstrated by integrating Portobello with the on-road simulator XR-OOM with 32 participants in both settings.

Honorable Mention Award at MobileHCI'23 - DOI: 10.1145/3604275
We delved into the impact of Infinite Scrolling on social media usage, particularly focusing on how it contributes to habitual and regretful use. Our study (N=46) defined and examined the 'loop' phenomenon that traps users in extended sessions and also emphasized the importance of considering the user's context in designing interventions, revealing that factors outside the app itself often influence breaks in social media usage.

Best Video Award at AutoUI'23 - DOI: 10.1145/3581961.3609854
I was honored to receive the Best Video Award, chosen by popular vote, for a video presentation in which we explored the nuanced distinctions between autonomous, automated, & highly automated vehicles. The video aimed to stimulate a balanced discussion on the potential impacts—positive, negative, & ambiguous—of truly autonomous vehicles that can act independently, possibly even against the wishes of their owners.

Honorable Mention Award at MobileHCI'22 - DOI: 10.1145/3546712
I was honored to receive an Honorable Mention Award for a full paper that developed a taxonomy of augmented reality visualizations for connected automated and manual driving, aiming to enhance trust in such systems. Our work, which included an evaluative driving simulator study, focused on how augmented reality can help drivers and passengers understand and trust the information provided by infrastructure-mounted sensors and onboard systems.

Honorable Mention Award at CHI'20 - DOI: 10.1145/3313831.3376472
We presented an inclusive user-centered design for vehicle-pedestrian communication to enhance the safety and experience of both vision-impaired and sighted pedestrians. Workshops with vision-impaired individuals and a virtual reality study revealed that communication from all relevant vehicles and detailed messaging significantly improves trust and understanding and reduces cognitive load.

SKILLS

- Coding: Programming in **R** (data analysis, visualization, web-scraping), Python (application of neural networks), Java (professionally at Airbus), & C# (especially with **Unity**)
- Research: **Quantitative Analysis** using R; experience with parametric and non-parametric data; linear regression and hierarchical models
- Research: **Qualitative Analysis** according to Saldaña
- Proficient in open science practices, encompassing transparency, reproducibility, & accessibility in research; experienced in sharing data (e.g, here) via open access platforms, utilizing open-source tools, contributing to community projects, & advocating for ethical research aligned with FAIR principles.
- Languages: German (native), English (proficient), French, Spanish (Intermediate), Latin
- Former Lifeguard with the Deutsche Lebens-Rettungs-Gesellschaft e.V. (DLRG; German Life Saving Association)

MAJOR PUBLICATIONS (CHI, IMWUT)

ACM CHI/IMWUT are widely recognized as the premier venues for publishing research in the field of HCI. They are highly competitive, with acceptance rates typically ranging between 20-25%.

CHI

1. P. Jansen*, **M. Colley***, S. Krauß, D. Hirschle, & E. Rukzio, OptCarVis: Improving Automated Vehicle Functionality Visualizations Using Bayesian Optimization to Enhance User Experience
   In Proc. of CHI 2025, ACM, doi: 10.1145/3706598.3713514, *Joint First Authors
   CHI Honorable Mention Award for Best Paper (top 5%)
2. **M. Colley***, P. Jansen*, M. Keskar, & E. Rukzio, Improving External Communication of Automated Vehicles Using Bayesian Optimization
   In Proc. of CHI 2025, ACM, doi: 10.1145/3706598.3714187, *Joint First Authors
3. **M. Colley**, J. Westhauser, J. Andersson, A. G. Mirnig, & E. Rukzio, Introducing ROADS: A Systematic Comparison of Remote Control Interaction Concepts for Automated Vehicles at Road Works
   In Proc. of CHI 2025, ACM, doi: 10.1145/3706598.3713476
4. M. Sasalovici, A. Zeqiri, R. C. Schramm, O. J. A. Nuñez, P. Jansen, J.P. Freiwald, **M. Colley**, C. Winkler, & E. Rukzio, Bumpy Ride? Understanding the Effects of External Forces on Spatial Interactions in Moving Vehicles
   In Proc. of CHI 2025, ACM, doi: 10.1145/3706598.371407
5. L. Meinhardt, L. Wilke, M. Elhaidary, J. von Abel, P. Fink, M. Rietzler, **M. Colley**, & E. Rukzio, Light My Way: Developing and Exploring a Multimodal Interface to Assist People With Visual Impairments to Exit Highly Automated Vehicles
   In Proc. of CHI 2025, ACM, doi: 10.1145/3706598.3713454
6. L. Meinhardt, C. Schramm, P. Jansen, **M. Colley**, & E. Rukzio, Fly Away: Evaluating the Impact of Motion Fidelity on Optimized User Interface Design via Bayesian Optimization in Automated Urban Air Mobility Simulations
   In Proc. of CHI 2025, ACM, doi: 10.1145/3706598.3713288
7. L. Meinhardt, M. Elhaidary, **M. Colley**, M. Rietzler, J. O. Rixen, A. K. Purohit, & E. Rukzio, Scrolling in the Deep: Analysing Contextual Influences on Intervention Effectiveness during Infinite Scrolling on Social Media
   In Proc. of CHI 2025, ACM, doi: 10.1145/3706598.3713187
8. **M. Colley**, O. Rajabi, & E. Rukzio, Investigating the Effects of External Communication and Platoon Behavior on Manual Drivers at Highway Access
   In Proc. of CHI 2024, ACM, doi: 10.1145/3613904.3642365

9. **M. Colley**, B. Wanner, M. Rädler, M. Rötzer, J. Frommel, T. Hirzle, P. Jansen, & E. Rukzio, Effects of a Gaze-Based 2D Platform Game on User Enjoyment, Perceived Competence, & Digital Eye Strain
In Proc. of CHI 2024, ACM, doi: 10.1145/3613904.3641909
10. D. Dey, T. Senan, B. Hengeveld, **M. Colley**, A. Habibovic, & W. Ju, Multi-Modal eHMIs: The Relative Impact of Light and Sound in AV-Pedestrian Interaction
In Proc. of CHI 2024, ACM, doi: 10.1145/3613904.3642031
11. F. Bu, S. Li, D. Goedicke, **M. Colley**, G. Sharma, & W. Ju, Portobello: Extending Driving Simulation from the Lab to the Road
In Proc. of CHI 2024, ACM, doi: 10.1145/3613904.3642341
CHI Honorable Mention Award for Best Paper (top 5%)
12. P. Jansen, J. Britten, A. Häusele, T. Segschneider, **M. Colley**, & E. Rukzio, AutoVis: Enabling Mixed-Immersive Analysis of Automotive User Interface Interaction Studies
In Proc. of CHI 2023, ACM, doi: 10.1145/3544548.3580760, [Website Link]
13. M. Lanzer, I. Koniakowsky, **M. Colley** and M. Baumann, Interaction Effects of Pedestrian Behavior, Smartphone Distraction and External Communication of Automated Vehicles on Crossing and Gaze Behavior
In Proc. of CHI 2023, ACM, doi: 10.1145/3544548.3581303
14. J. O. Rixen, **M. Colley**, A. Askari, J. Gugenheimer, & E. Rukzio, Consent in the Age of AR: Investigating The Comfort With Displaying Personal Information in Augmented Reality
In Proc. CHI 2021, ACM, doi: 10.1145/3491102.3502140
15. **M. Colley**, E. Bajrovic, & E. Rukzio, Effects of Pedestrian Behavior, Time Pressure, & Repeated Exposure on Crossing Decisions in Front of Automated Vehicles Equipped with External Communication
In Proc. CHI 2022, ACM, doi: 10.1145/3491102.3517571
16. J. O. Rixen, T. Hirzle, **M. Colley**, Y. Etzel, E. Rukzio and J. Gugenheimer, Exploring Augmented Visual Alterations in Interpersonal Communication
In Proc. CHI 2021, ACM, doi: 10.1145/3411764.3445597
17. **M. Colley**, B. Eder, J. O. Rixen, & E. Rukzio, Effects of Semantic Segmentation Visualization on Trust, Situation Awareness, & Cognitive Load in Highly Automated Vehicles
In Proc. CHI 2021, ACM, doi: 10.1145/3411764.3445351
18. K. Holländer*, **M. Colley***, E. Rukzio and A. Butz, A Taxonomy of Vulnerable Road Users for HCI Based On A Systematic Literature Review
In Proc. CHI 2021, ACM, doi: 10.1145/3411764.3445480
19. **M. Colley**, M. Walch, J. Gugenheimer, A. Askari, & E. Rukzio, Towards Inclusive External Communication of Autonomous Vehicles for Pedestrians with Vision Impairments
In Proc. CHI 2020, ACM, doi:, 10.1145/3313831.3376472
CHI Honorable Mention Award for Best Paper (top 5%)
20. K. Rogers, **M. Colley**, D. Lehr, J. Frommel, M. Walch, L. E. Nacke and M. Weber, KickAR: Exploring Game Balancing Through Boosts and Handicaps in Augmented Reality Table Football
In Proc. CHI 2018, ACM, doi: 10.1145/3173574.3173740

IMWUT

1. **M. Colley***, S. Hartwig*, A. Zeqiri, T. Ropinski, & E. Rukzio, AutoTherm: A Dataset and Benchmark for Thermal Comfort Estimation Indoors and in Vehicles
In Proc. IMWUT 2024, ACM, *Joint First Authors, doi: 10.1145/3678503
2. A. Stampf, **M. Colley**, A. Knuth, C. Tasci, & E. Rukzio, Examining Psychological Conflict-Handling Strategies for Highly Automated Vehicles to Resolve Legal User-Vehicle Conflicts
In Proc. IMWUT 2024, ACM, doi: 10.1145/3678511
3. A. Stampf, M. Sasalovici, L. Meinhardt, **M. Colley**, M. Giss, & E. Rukzio, Move, Connect, Interact: Introducing a Design Space for Cross-Traffic Interaction
In Proc. IMWUT 2024, ACM, doi: 10.1145/3678580
4. L. Meinhardt, M. Rück, J. Zähnle, M. Elhaidary, **M. Colley**, M. Rietzler, & E. Rukzio, Hey, What's Going On? Conveying Traffic Information to People with Visual Impairments in Highly Automated Vehicles: Introducing OnBoard
In Proc. IMWUT 2024, ACM, doi: 10.1145/3659618
5. **M. Colley**, O. Speidel, J. Strohbeck, J. O. Rixen, J. H. Belz, & E. Rukzio, Effects of Uncertain Trajectory Prediction Visualization in Highly Automated Vehicles on Trust, Situation Awareness, & Cognitive Load
In Proc. IMWUT 2023, ACM, doi: 10.1145/3631408, [Video Link]
6. **M. Colley***, L. Meinhardt*, A. Fassbender, M. Rietzler, & E. Rukzio, Come Fly With Me - Investigating the Effects of Path Visualizations in Automated Urban Air Mobility
In Proc. IMWUT 2023, ACM, doi: 10.1145/3596249, *Joint First Authors
7. **M. Colley**, J. Britten, & E. Rukzio, Scalability in External Communication of Automated Vehicles: Evaluation and Recommendations
In Proc. IMWUT 2023, ACM, doi: 10.1145/3596248
8. P. Jansen, **M. Colley**, & E. Rukzio, A Design Space for Human Sensor and Actuator Focused In-Vehicle Interaction Based on a Systematic Literature Review
In Proc. IMWUT 2022, ACM, doi: 10.1145/3534617
9. **M. Colley**, M. Rädler, J. Glimmann, & E. Rukzio, Effects of Scene Detection, Scene Prediction, & Maneuver Planning Visualizations on Trust, Situation Awareness, & Cognitive Load in Highly Automated Vehicles
In Proc. IMWUT 2022, ACM, doi: 10.1145/3534609, [Video Link]
10. **M. Colley**, P. Jansen, E. Rukzio and J. Gugenheimer, SwiVR-Car-Seat: Exploring Vehicle Motion Effects on Interaction Quality in Virtual Reality Automated Driving Using a Motorized Swivel Seat
In Proc. IMWUT 2021, ACM, doi: 10.1145/3494968
11. **M. Colley**, S. Krauß, M. Lanzer, & E. Rukzio, How Should Automated Vehicles Communicate Critical Situations? A Comparative Analysis of Visualization Concepts
In Proc. IMWUT 2021, ACM, doi: 10.1145/3478111

FURTHER PUBLICATIONS

Journal paper
Transportation Research Part F, with an impact factor of 4.60 (2022), and the International Journal of Human-Computer Studies, with an impact factor of 5.40 (2024), are considered to be top-tier journals in traffic psychology.

1. P. Jansen*, **M. Colley***, T. Pfeifer, & E. Rukzio, Visualizing Imperfect Situation Detection and Prediction in Automated Vehicles: Understanding Users
In Transportation Research Part F: Traffic Psychology and Behaviour 2024, Elsevier, *Joint First Authors, doi: 10.1016/j.trf.2024.05.015
2. A. Stampf, A. Knuth, **M. Colley**, & E. Rukzio, Law and Order: Investigating the Effects of Conflictual Situations in Manual and Automated Driving in a German Sample
In International Journal of Human-Computer Studies 2024, Elsevier, doi: 10.1016/j.ijhcs.2024.103260

3. **M. Colley**, B. Wankmüller, T. Mend, T. Väth, E. Rukzio and J. Gugenheimer, User Gesticulation Inside an Autonomous Vehicle with External Communication can Cause Confusion in Pedestrians and a Lower Willingness to Cross
In Transportation Research Part F: Traffic Psychology and Behaviour 2022, Elsevier, doi: 10.1016/j.trf.2022.03.011
4. **M. Colley**, C. Hummler, & E. Rukzio, Effects of Mode Distinction, User Visibility, & Vehicle Appearance on Mode Confusion When Interacting With Highly Automated Vehicles
In Transportation Research Part F: Traffic Psychology and Behaviour 2022, Elsevier, doi: 10.1016/j.trf.2022.06.020
5. **M. Colley**, S. Mytilineos, M. Walch, E. Rukzio, & J. Gugenheimer, Requirements for the Interaction With Highly Automated Construction Site Delivery Trucks
In Proc. Frontiers 2022, Frontiers, doi: 10.3389/fhumd.2022. 794890
6. M. Lanzer, T. Stoll, **M. Colley**, & M. Baumann, Intelligent Mobility in the City: The Influence of System and Context Factors on Drivers' Takeover Willingness and Trust in Automated Vehicles
In Proc. Frontiers 2021, Frontiers, doi: 10.3389/fhumd.2021.676667

Conference full paper

1. P. Fink, H. Milne, A. Caccese, M. Alsamsam, J. Loranger, **M. Colley**, N. Giudice, Accessible Maps for the Future of Inclusive Ridesharing
In Proc. AutoUI 2024, ACM, doi: 10.1145/3640792.3675736
2. A. Asha, **M. Colley**, S. Sultana, E. Sharlin, Introducing AV-Sketch: An Immersive Participatory Design Tool for Automated Vehicle — Passenger Interaction
In Proc. AutoUI 2024, ACM, doi: 10.1145/3640792.3675705
3. P. Ebel, P. Bazilinskyy, **M. Colley**, C. Goodridge, P. Hock, C. Janssen, H. Sandhaus, A. Srinivasanm, P. Wintersberger, Changing Lanes Toward Open Science: Openness and Transparency in Automotive User Research
In Proc. AutoUI 2024, ACM, doi: 10.1145/3640792.3675730
4. A. Stampf, **M. Colley**, B. Girst, E. Rukzio, Exploring Passenger-Automated Vehicle Negotiation Utilizing Large Language Models for Natural Interaction
In Proc. AutoUI 2024, ACM, doi: 10.1145/3640792.3675725
5. A. Stampf* and **M. Colley***, Deriving Non-Driving-Related Activities in Highly Automated Driving via an Autoethnographic Approach by Traveling Canada in a Recreational Vehicle
In Proc. MuC 2024, *Joint First Authors, doi: 10.1145/3670653.3670663
6. T. Wagner*, **M. Colley***, D. Breckel, M. Kösel, & E. Rukzio, UnitEye: Introducing a User-Friendly Plugin to Democratize Eye Tracking Technology in Unity Environments
In Proc. MuC 2024, *Joint First Authors, doi: 10.1145/3670653.3670655,
7. **M. Colley***, P. Jansen*, J. J. Matthiesen*, H. Hoberg, C. Reger, & I. Thiermann, How Much Home Office is Ideal? A Multi-Perspective Algorithm
In Proc. CHIWORK 2023, ACM, doi: 10.1145/3596671.3596672, *Joint First Authors
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10. J. O. Rixen, L. Meinhardt, M. Glöckler, A. Schlothauer, M. Ziegenbein, **M. Colley**, J. Gugenheimer, & E. Rukzio, The Loop and Reasons to Break It: Investigating Infinite Scrolling Behaviour in Social Media Applications and Reasons to Stop
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11. S. Suzuki, **M. Colley**, S. Li, I. Mandel, A. Stampf and W. Ju, AdVANcing Design: Customizing Spaces for Vanlife
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13. **M. Colley**, J. O. Rixen, W. I. Pellegrino, & E. Rukzio, (Eco-)Logical to Compare? - Utilizing Peer Comparison to Encourage Ecological Driving in Manual and Automated Driving
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Conference short paper

1. **M. Colley**, J. Czymmeck, P. Jansen, L.-M. Meinhardt, P. Ebel, & E. Rukzio, UAM-SUMO: Simulacra of Urban Air Mobility Using SUMO To Study Large-Scale Effects
In Proc. HRI 2025, ACM, doi: 10.5555/3721488.3721610, GitHub Link
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Workshops

1. M. Haimerl, P. Jansen, A. Riener and **M. Colley**, Accessible Automated Automotive Workshop Series (A3WS): Accessibility in Mobility
In Proc. MuC EA 2025, accepted, Gesellschaft für Informatik e.V, Website: https://a3ws.github.io/MuC2025/
2. C. Parker, S. Yoo, J. Fredericks, T.T.M. Tran, **M. Colley**, Y. Lee, K. Le, S. Stannus, W. Woo, M. Billinghurst, The Third Workshop on Building an Inclusive and Accessible Metaverse for All
In Proc. CHI EA 2025, ACM, doi: 10.1145/3706599.3706730, Website: https://sites.google.com/view/accessiblemetaverse
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4. Y. W. Kim, Y. G. Ji, S. H. Yoon, **M. Colley** and L. Meinhardt, The 3rd Workshop on User Experience in Mobility: What Could We Learn From AutomotiveUI?
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6. A. Löcken, A. Matviienko, **M. Colley**, D. Dey, A. Habibovic, Y. M. Lee and A. Riener, Accessible Automated Automotive Workshop Series (A3WS): International Perspective on Inclusive External Human-Machine Interfaces
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Extended abstracts

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10. **M. Colley**, M. Walch, & E. Rukzio, For a Better (Simulated) World: Considerations for VR in External Communication Research
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Demos

- L. Meinhardt*, **M. Colley***, A. Fassbender, & E. Rukzio, Stairway to Heaven: A Demonstration of Different Trajectories and Weather Conditions in Automated Urban Air Mobility
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- P. Jansen, J. Britten, A. Häusele, T. Segschneider, **M. Colley**, & E. Rukzio, A Demonstration of AutoVis: Enabling Mixed-Immersive Analysis of Automotive User Interface Interaction Studies
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Videos

- R. Bernhaupt, **M. Colley**, D. Goedicke, A. Meschtscherjakov, B. Pfleging, A. Riener, & S. Sadeghian, A Critical Perspective on Radically Innovating Personal Mobility
  In Proc. AutoUI EA 2022, ACM, doi: 10.1145/3544999.3551689, [Video Link]
- **M. Colley**, S. Li, B. P. V. Samsom, & D. Sogemeier, What If Automated Vehicles Became AUTONOMOUS? A Critical Perspective
  In Proc. AutoUI EA 2023, ACM, doi: 10.1145/3581961.3609854, [Video Link]
  AutoUI Best Video Award (People's Choice)

Workshop position papers

1. **M. Colley**, & E. Rukzio, Challenges of Explainability, Cooperation, & External Communication of Automated Vehicles at CHI 2022
2. M. Walch, **M. Colley**, P. Hock, E. Rukzio, & M. Weber, Turn Drivers Into Users and Keep Them Out-Of-The-Loop to Save Energy at CHI 2020
3. **M. Colley**, M. Walch, & E. Rukzio, Towards Reducing Energy Waste through Usage of External Communication of Autonomous Vehicles at CHI 2020